The Quine-Mc-Cluskey Method

Also known as the method of PI.

→ When there are five variables, the k-map method is difficult to apply and completely impractical beyond five.

→ The Quine-McCluskey method is a formal tabular method for applying the Booleam distribibutive law to various terms to minimum sum of products by eliminating literal that appear in two terms as complements.

It is alternative of k-maps.

It is based on the concept of prime implicants.

Advantage: It is systematic for producing a minimal function that is less dependent on visual patterns.

Tabular Method

Simplify the following boolean expression.

$Y(A,B,C,D) = \Sigma m(0,1,3,7,8,9,11,15)$

Table-1

X	ABCD
0	0000
1	0001
3	0011
7	0111
8	1000
9	1001
11	1011
15	1111

No of 1's	Minterm	ABCD
0	0	0000
1	1	0001
	8	1000
2	3	0011
	9	1001
3	7	0111
	11	1011
4	15	1111

Table-2

Disadvantage:

Computational complexity still remains high.

→ The basic principle is to remove the terms, which are redundant and can be obtained by other terms.

(Table-2)

No of 1's	Minterm	1st level
0	m0, m1	000X
	m0, m8	X000 ✓
	m0, m2	00X0 ✓
1	m2, m10	X010 ✓
	m8, m10	10X0 ✓
	m8, m11	10X0 ✓
2	m10, m11	101X ✓
3	m11, m15	1X11 ✓
	m14, m15	111X ✓

Table-3

No of 1's	Minterm	2nd level	Binary e
0	m0, m8, m2, m10	X0X0	$\bar{B}\bar{D}$
	m0, m2, m8, m10	X0X0	
2	m10, m11, m11, m15	1X1X	AC
	m11, m15, m10, m11	1X1X	

PI	Minterm	m0	m1	m2	m8	m10	m11	m14	m15
$\bar{B}\bar{D}$	0,8,2,10	✓		✓	✓	✓			
AC	10,11,11,15					✓	✓		✓
$\bar{A}\bar{B}\bar{C}$	0,1	✓	✓						

$$F(A,B,C,D) = \bar{B}\bar{D} + AC + \bar{A}\bar{B}\bar{C}$$

Tabula بنتے Don't care W.D سے ✓ لگاتے وقت

Don't care term کو include نہیں کرنا۔

K-maps:

for a:

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$F(ABCD) = \bar{B}\bar{D} + C + BD + A$

for b

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	X	X	X	X
10	1	1	X	X

$F(ABCD) = \bar{B} + \bar{C}\bar{D} + CD$

for c

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	X	X	X	X
10	1	1	X	X

$F(ABCD) = \bar{C} + D + B$

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	X	X	X	X
10	1	1	X	X

$F(ABCD) = \bar{B}\bar{D} + \bar{B}C + B\bar{C}D + \bar{C} + A$

for e

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	X	X	X	X
10	1	0	X	X

$F(ABCD) = \bar{B}\bar{D} + C\bar{D}$

for g

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$F(ABCD) = \bar{C}\bar{D} + B\bar{C} + B\bar{D} + A$

for g

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

$F(ABCD) = \bar{B}C + B\bar{C} + A + B\bar{D}$